Inspiring Innovation and Leadership

KARATINA UNIVERSITY

SPECIAL/SUPPLEMENTARY EXAMINATIONS
2023/2024 ACADEMIC YEAR

THIRD YEAR **SECOND** SEMESTER EXAMINATION

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY/ BACHELOR OF SCIENCE IN COMPUTER SCIENCE

COURSE CODE: COM 311

COURSE TITLE: COMPUTER GRAPHICS

DATE: 17TH July 2024 **TIME: 12PM-2PM**

INSTRUCTION TO CANDIDATES

- SEE INSIDE

INSTRUCTIONS:

Answer Question 1 and any other 2 questions

Show workings for all computations

QUESTION ONE (30 MARKS)

a. Differentiate between Pixels and Texels as used in computer graphics. [2 marks]

b. Differentiate between the following terms as used in computer graphics

   i. Reflection as a 2D transformation and rotation as a 2D transformation [2 marks]

   ii. Diffuse reflection and specular reflection. [2 marks]

c. Goroud shading results to rough looking images in appearance as compared to phong shading. Elaborate on this statement. [4 marks]

d. Describe the relationship between the framebuffer and rasterization. [4 marks]

e. Epxlain any THREE features of OpenGL [6 marks]

f. Use a diagram to illustrate the architecture of OpenGL. [6 marks]

g. Describe the most appropriate type of projections to apply for a computer aided design tool used by an architect. [2 marks]

h. State the purpose glMatrixMode() in OpenGL camera projections. [2 marks]

QUESTION TWO (20 MARKS)

a. Explain THREE advantages of WebGL compared to plain OpenGL. [6 marks]

b. Given that a line is expressed by the equitation y=mx+c, assume two points a(x0,y0)and b(x1,y1). Derive the equation for y and m and write an algorithm for a computer programme that will plot a line. [8 marks]

QUESTION THREE (20 MARKS)

a. Describe any TWO tasks carried out by a dedicated graphics processor. [4 Marks]

b. Differentiate between the following terms as used in computer graphics:

i. interlaced and non-interlaced displays [2 marks]
ii. Physical and Synthetic images as used in computer graphics. [2 marks]
iii. GLU and GLUT components of an OpenGL Library [2 marks]

c. A point C on a sphere is placed such that the light bouncing on it is reflected at 90 degrees. Given that A and B are two vectors parallel to the surface of the sphere at point C, compute the surface normal at point C. [10 marks]

QUESTION FOUR (13 MARKS)

An image has the following coordinates:

$$A = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$$

Required:

a. Compute the new coordinates of the points in the image if the image is zoomed out by a factor of 2. [4 marks]

b. Compute the new coordinates of the points if the image is reflected along the y-axis. [4 marks]

QUESTION FIVE (20 MARKS)

a. Explain the term refresh as used in Computer Graphics. [2 marks]

b. Write a HTML code that creates a 600px by 300px canvas. [4 marks]

c. Write WebGL code that contextualizes the canvas in (c) above [6 marks]

d. Write code that displays a rectangle filled with the colour green. [8 marks]